

Splunk Universal Forwarder Windows Event Log Collection Lab

This lab demonstrates configuring centralized Windows event log collection from a Windows Server 2022 Active Directory environment into Splunk Enterprise running on an Ubuntu Docker host using Splunk Universal Forwarder.

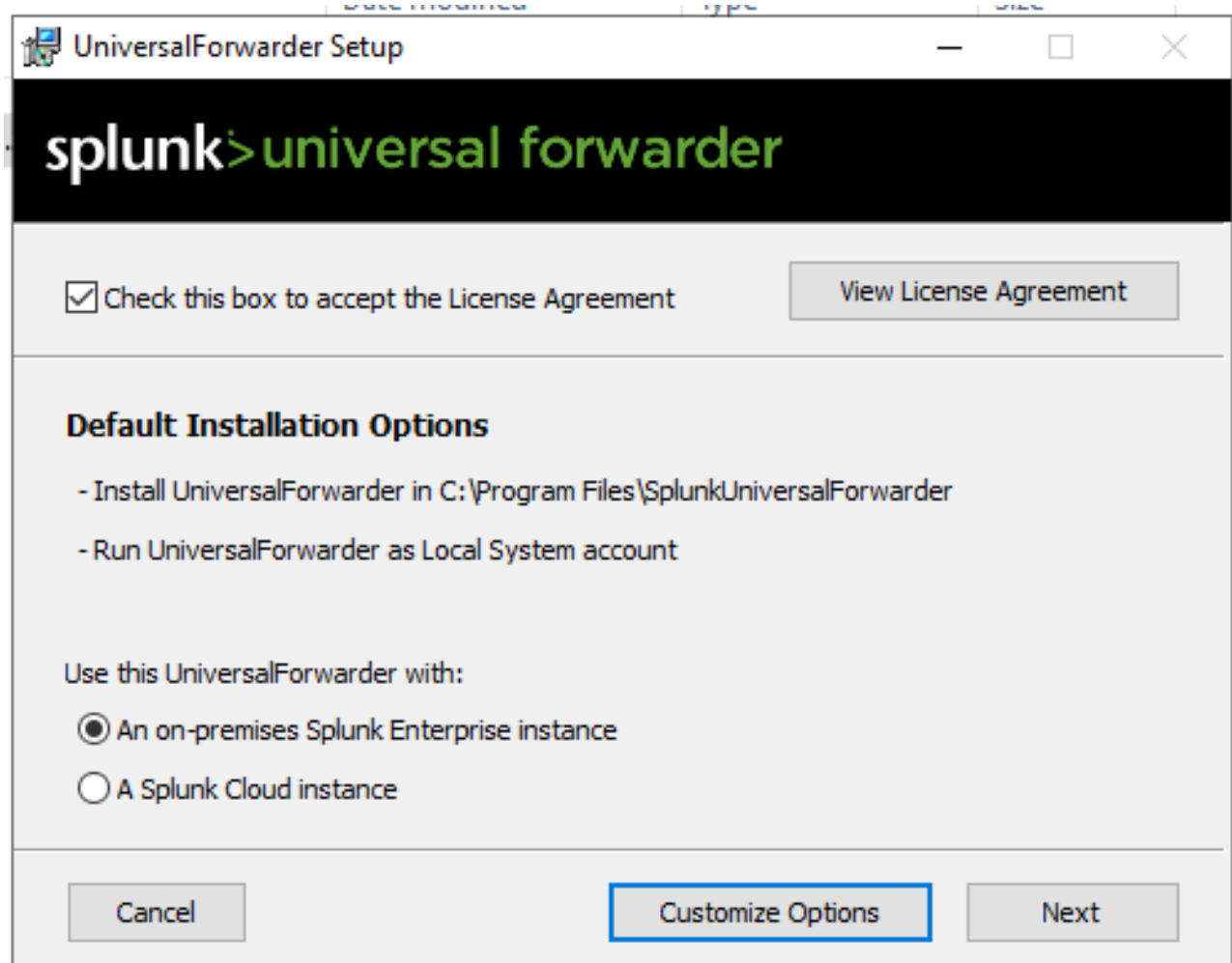
Lab Environment

Component	Details
SIEM Platform	Splunk Enterprise (Docker Container)
Forwarder	Splunk Universal Forwarder 10.2.3
Server OS	Windows Server 2022
Host OS	Ubuntu Linux
Virtualization	VMware Workstation
Log Transport Port	TCP 9997

Objective: Configure centralized Windows event log forwarding from a Windows Server 2022 domain controller to Splunk Enterprise using Splunk Universal Forwarder and validate successful log ingestion.

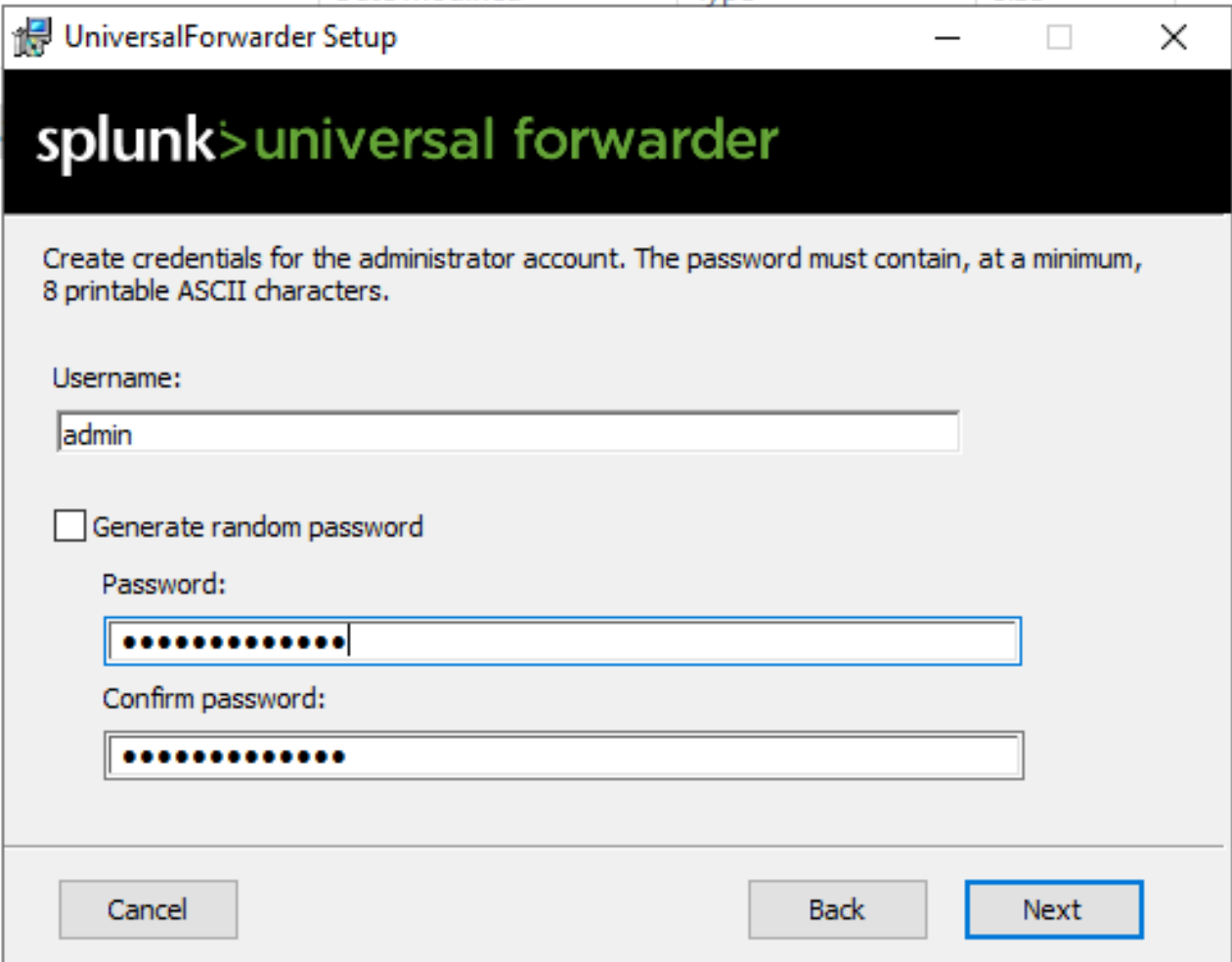
Step 1 - Download and Install Splunk Universal Forwarder

The Splunk Universal Forwarder installer was launched on the Windows Server 2022 virtual machine.



Step 2 - Configure Universal Forwarder Administrative Credentials

Administrative credentials for the forwarder service were configured during installation.



The image shows a Windows-style window titled "UniversalForwarder Setup". The window has a black header bar with the "splunk" logo in white and "universal forwarder" in green. Below the header, there is a light gray area with the text: "Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters." Below this text, there are three input fields: "Username:" with the value "admin", "Password:" with 10 black dots, and "Confirm password:" with 10 black dots. To the left of the password fields is a checkbox labeled "Generate random password" which is unchecked. At the bottom of the window, there are three buttons: "Cancel", "Back", and "Next". The "Next" button is highlighted with a blue border.

UniversalForwarder Setup

splunk>universal forwarder

Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.

Username:

admin

☐ Generate random password

Password:

Confirm password:

Cancel Back Next

Step 3 - Configure Receiving Indexer Connection

The Ubuntu-based Splunk Enterprise server IP address and receiving port 9997 were configured as the receiving indexer.

UniversalForwarder Setup

splunk>universal forwarder

If you intend to use a Splunk receiving indexer to configure this UniversalForwarder, please specify the host or IP, and port (default port is 9997). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

Receiving Indexer

Hostname or IP

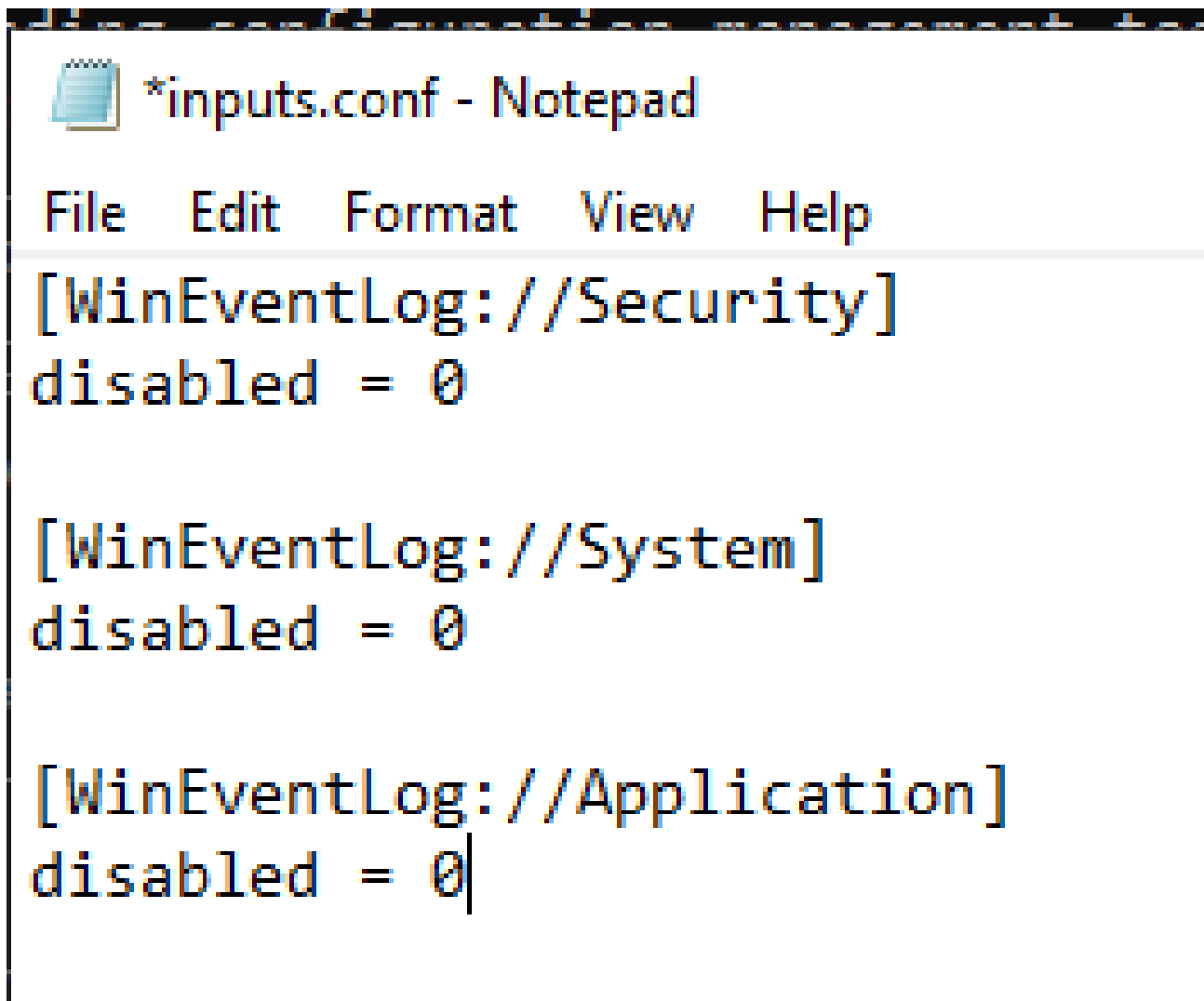
192.168 : 9997

Enter the hostname or IP of your receiving indexer, e.g. ds.splunk.com *default is 9997*

Cancel Back Next

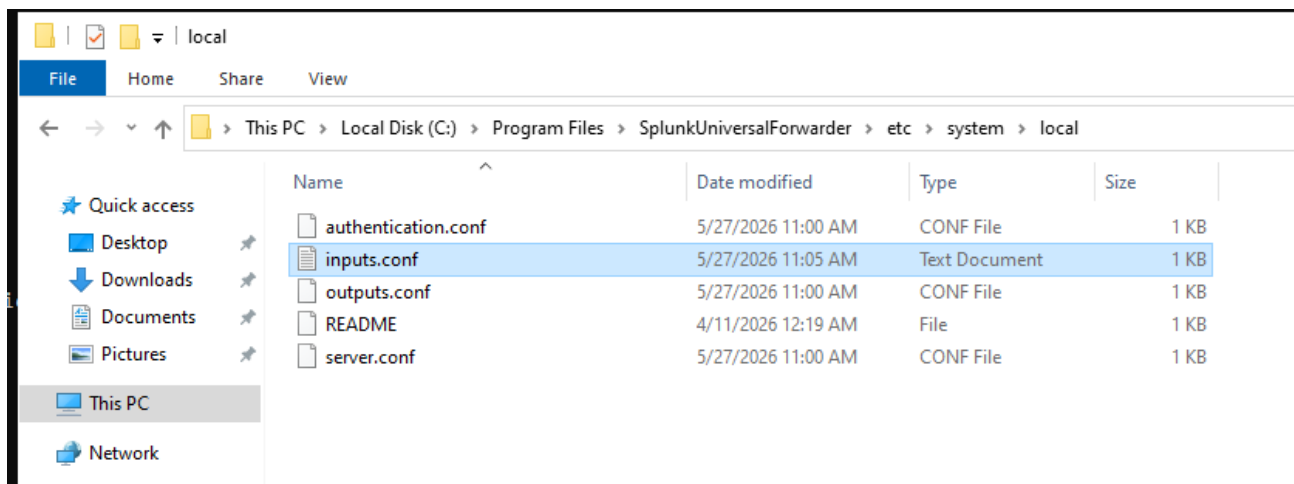
Step 4 - Configure Windows Event Log Collection

Windows Event Log collection was enabled through the inputs.conf configuration file.



Step 5 - Save *inputs.conf* to Splunk Universal Forwarder Local Directory

The *inputs.conf* file was saved to the Splunk Universal Forwarder local configuration directory.



Step 6 - Restart Splunk Universal Forwarder Service

The Splunk Universal Forwarder service was restarted to apply configuration changes.

```
C:\Program Files\SplunkUniversalForwarder\bin>. \splunk restart
SplunkForwarder: Stopped

Splunk> Needle. Haystack. Found.

Checking prerequisites...
  Checking mgmt port [REDACTED]: open
New certs have been generated in 'C:\Program Files\SplunkUniversalForwarder\etc\auth'.
  Checking conf files for problems...
  Done
  Checking default conf files for edits...
  Validating installed files against hashes from 'C:\Program Files\SplunkUniversalForwarder\splunkforwarder-10.2.3-4d61cf8a5c0c-windows-x64-manifest'
  All installed files intact.
  Done
All preliminary checks passed.

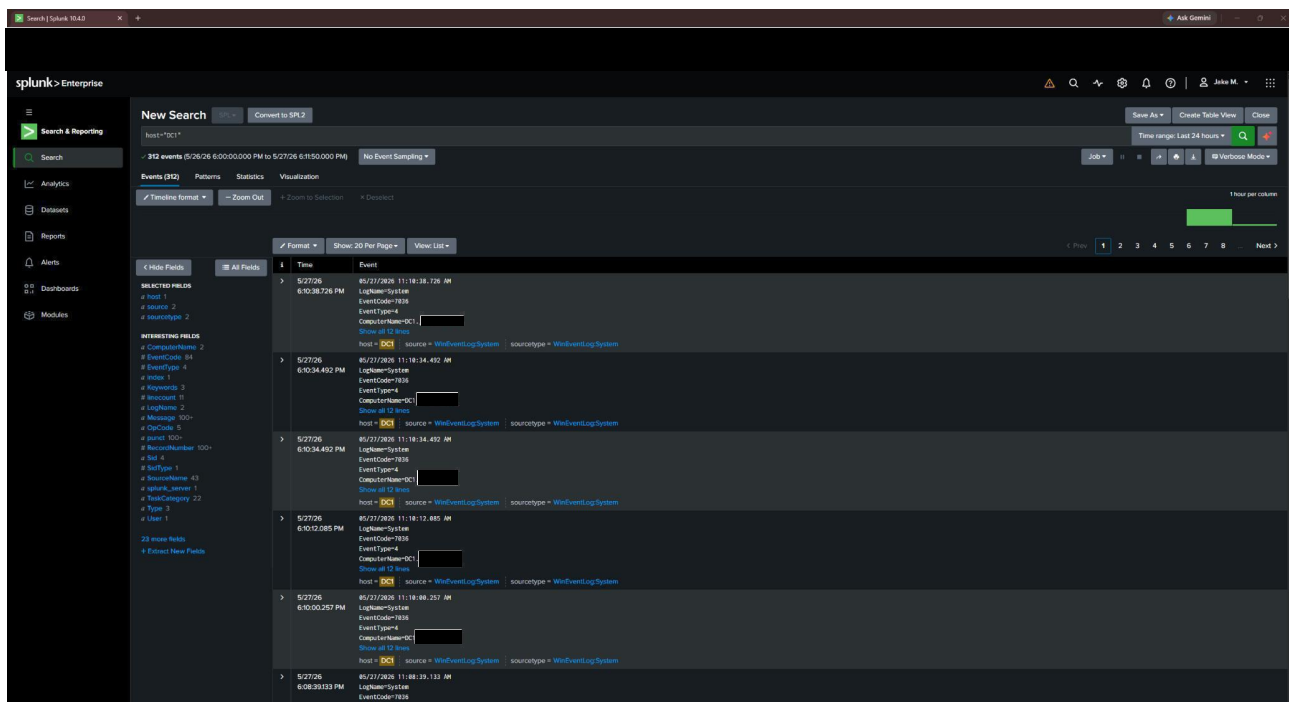
Starting splunk server daemon (splunkd)...

SplunkForwarder: Starting (pid 6828)
Done

C:\Program Files\SplunkUniversalForwarder\bin>
```

Step 7 - Verify Windows Event Logs in Splunk Enterprise

Windows event logs were successfully received and indexed within Splunk Enterprise.



Challenges Encountered

- Troubleshooting VMware network adapter connectivity for communication between the Windows Server VM and Ubuntu host.
- Configuring the correct receiving indexer IP address and TCP forwarding port.
- Manually creating and validating the inputs.conf file for Windows Event Log collection.
- Restarting the Splunk Universal Forwarder service to ensure configuration changes were applied correctly.

Conclusion

This lab successfully demonstrated centralized Windows event log collection using Splunk Universal Forwarder and Splunk Enterprise in a virtualized Active Directory environment. The project provided hands-on experience with SIEM deployment, Windows event monitoring, log forwarding, configuration management, and troubleshooting enterprise-style logging infrastructure.